

SLU INTERNATIONAL STUDENT COMPLIANCE ANALYTICS

FINAL PRESENTATION — WEEK 4

SAINT LOUIS UNIVERSITY | JUNE 2026

TEAM 25

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PROJECT OVERVIEW

What This Project Is About

SLU's Office of International Services manages thousands of F-1 international students. They need to make sure every student has paid their SEVIS fee, has proper documents, and is tracked correctly in the system.

Our job was to take their raw data, clean it up, build a dashboard, and find where the real compliance problems are hiding.

Simple questions we tried to answer:

- Who hasn't paid their SEVIS fee?
- Which students are missing documents?
- How many applicants are stuck in the pipeline?
- Where is the data broken or incomplete?

DATASETS OVERVIEW

Datasets Used

Dataset 1 — SEVIS Export (Federal DHS System)

- Records: 3,524 F-1 Initial Status Students
- Raw Columns: 126 | Cleaned Columns: 81
- Covers: Identity, academic programme, financial sponsorship, travel documents, I-901 fee status

Dataset 2 — Connect / Applicant (SLU Internal Workflow System)

- Records: 6,875 unique applicants (raw rows: 34,341 before deduplication)
- Covers: I-20 issuance pipeline, deposit payment status, advisor assignment

Dataset 3 — Google Sheets Copy of Connect

- VERDICT: EXCLUDED
- Reason: Duplicate of Connect CSV with degraded timestamp precision — no unique analytical value

Tools Used:

Excel | Supabase (PostgreSQL) | pgAdmin 4 | Looker Studio (Data Studio)

WEEK 1: CLEANING THE DATA

Honestly, the data was a mess when we started.

SEVIS Dataset Issues We Fixed:

- Removed 45 useless columns (empty or all same value)
- Dates were stored as Unix timestamps — we decoded them properly
- Some amount columns had text errors from Excel formatting
- Standardized everything to consistent formats

Applicant Dataset Issues We Fixed:

- Found the same data repeated 5 times — removed 27,466 duplicate rows
- Two different date formats mixed together — unified them
- Fixed column name typos
- Dropped 19 redundant columns

Biggest Problem We Found:

There is no common field between SEVIS and Applicant data. You can't connect them. OIS needs to provide a mapping table to fix this — without it, full compliance reporting isn't possible.

WEEK 1: DATA GAPS THAT MATTER

BEFORE WE EVEN STARTED ANALYSIS, WE FOUND SERIOUS GAPS:

- 53% OF STUDENTS — NO SEVIS FEE PAYMENT ON RECORD
- 91% OF STUDENTS — NO PASSPORT OR VISA INFO ON FILE
 - 48% OF STUDENTS — NO FINANCIAL DATA ON THEIR I-20
 - 57.6% OF APPLICANTS — NO ADVISOR ASSIGNED
- NO WAY TO CONNECT SEVIS DATA WITH APPLICANT DATA

THESE AREN'T SMALL ISSUES. THE FEE GAP ALONE MEANS OVER HALF THE STUDENT BODY TECHNICALLY CAN'T GET A US VISA RIGHT NOW.

WEEK 2: WHAT THE DATA ACTUALLY SHOWED (ELIGIBILITY)

Eligibility Analysis Results:

Total students analyzed: 3,524

Countries represented: 66

Students who met eligibility requirements: 1,602 (45%)

What stood out:

- Students who submitted documents early were consistently more compliant across all categories
 - India is by far the largest group — 59% of all students
- That's a concentration risk if anything delays Indian visa processing
- Gender split: roughly 66% Male, 34% Female
 - Master's level students make up the largest education group

The 45% eligibility rate tells us more than half the students still have something incomplete or unresolved.

DOCUMENT & FEE PAYMENT ANALYSIS

Document Compliance:

- Only 311 students have a passport number on file
- Only 312 have a visa number recorded
- 1,672 have a US entry date — meaning they already arrived
- The rest are pre-arrival — so missing documents is expected

Fee Payment (I-901 SEVIS Fee):

- Total fees collected: \$360,700
- 53 fee cancellations recorded
- India paid the most fees by far
- 53% of students have NO payment record at all

This 53% gap is the single biggest compliance risk in the project. A student cannot get a US visa without paying this fee.

WEEK 3: THE DASHBOARD WE BUILT

We built a 7-tab live dashboard in Looker Studio connected directly to our PostgreSQL database.

Tab 1 & 2 → Who the students are (eligibility, countries, majors)

Tab 3 & 4 → Document gaps (passport, visa, I-94, I-20 reasons)

Tab 5 → Government fee payment tracking

Tab 6 → Applicant pipeline (deposits, I-20s, advisors)

Tab 7 → Summary of everything on one page

We also ran SQL queries in pgAdmin to double-check every number the dashboard showed. Most matched perfectly. Two discrepancies were found and flagged for OIS to investigate.

THE 5 BIGGEST RISKS WE FOUND

1. SEVIS Fee Gap

Over half the students (53%) have no fee payment on record. This is not a data error — it is a genuine compliance emergency.

2. India Concentration

59% of students are from India. One visa processing disruption there affects the majority of SLU's international student body.

3. Pipeline Bottleneck

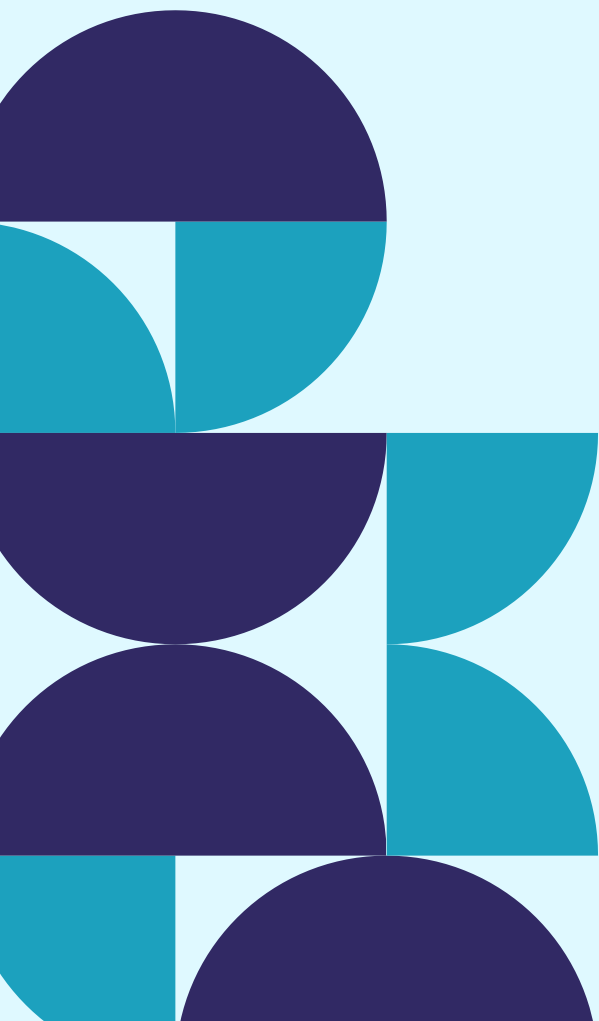
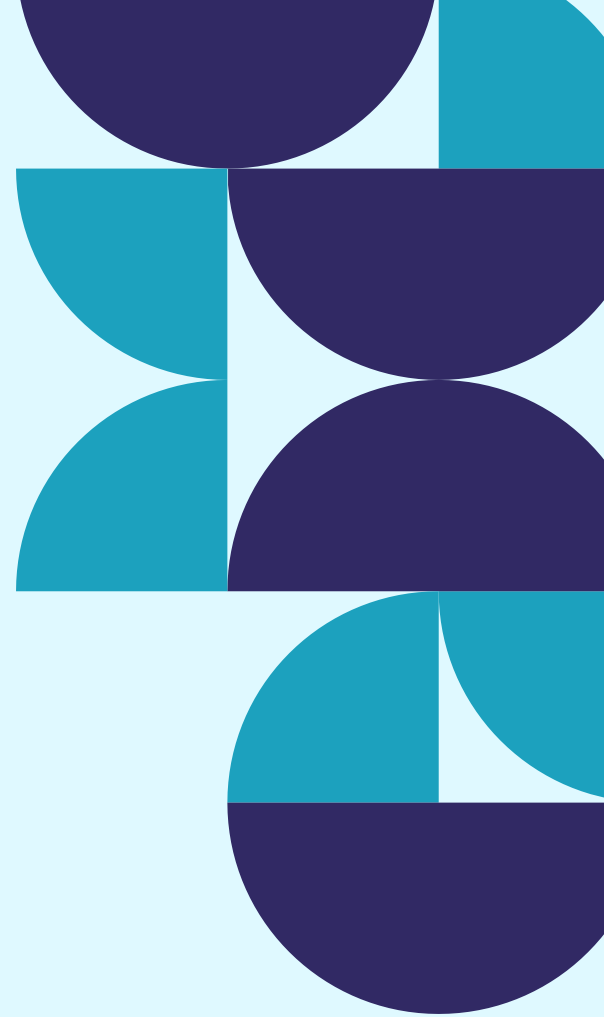
6,875 applicants in the system. Only 1,512 paid their deposit. 3,962 have no advisor assigned — they're just sitting there.

4. Document Labels Are Misleading

91% "missing" passport/visa sounds alarming but most students haven't arrived yet. The dashboard label needs to say "pre-arrival" not "missing."

5. No Link Between Datasets

SEVIS and Applicant data cannot be joined without a mapping table. Right now OIS cannot see a student's full journey from application to SEVIS in one place.



OUR RECOMMENDATIONS

WHAT OIS SHOULD DO NEXT:

FEE PAYMENT

- SEND ALERTS TO THE 1,869 STUDENTS WITH NO PAYMENT RECORD
- BUILD A WEEKLY REPORT SHOWING WHO IS APPROACHING VISA DEADLINES WITHOUT HAVING PAID

PIPELINE MANAGEMENT

- ASSIGN ADVISORS TO ALL 3,962 UNASSIGNED APPLICANTS IMMEDIATELY
- INVESTIGATE THE 877 CASES WHERE I-20 WAS ISSUED BEFORE DEPOSIT

DATA & DASHBOARD

- FIX THE MISLEADING "CANCELLATION" LABEL FOR MISSING FEE RECORDS
 - GET THE SEVIS ↔ APPLICANT MAPPING TABLE FROM OIS IT TEAM
- KEEP THIS DASHBOARD RUNNING AS A PERMANENT MONITORING TOOL

LONG TERM

- SET A FIXED EXPORT SCHEDULE SO DUPLICATE SNAPSHOTS STOP APPEARING
 - STANDARDIZE DATE FORMATS AT THE SOURCE SYSTEM LEVEL

CONCLUSION

WHAT OUR TEAM DELIVERED IN 4 WEEKS:

- ✓ CLEANED AND LOADED 3 DATASETS INTO A LIVE DATABASE
 - ✓ REMOVED 27,466 DUPLICATE ROWS FROM RAW DATA
 - ✓ FIXED 6 TYPES OF DATA CORRUPTION ISSUES
- ✓ BUILT A 7-TAB COMPLIANCE DASHBOARD IN LOOKER STUDIO
 - ✓ VALIDATED EVERY DASHBOARD METRIC USING SQL
- ✓ IDENTIFIED AND DOCUMENTED 5 REAL COMPLIANCE RISKS

THE DASHBOARD OIS NOW HAS ISN'T JUST A CHART — IT'S A TOOL THEY CAN OPEN EVERY MORNING AND IMMEDIATELY SEE WHICH STUDENTS NEED ATTENTION BEFORE IT BECOMES A LEGAL OR VISA PROBLEM.

THAT WAS THE WHOLE POINT OF THIS PROJECT.

VIDEO RECORDING

▶ [Click here to watch the recording](#)

THANK YOU

